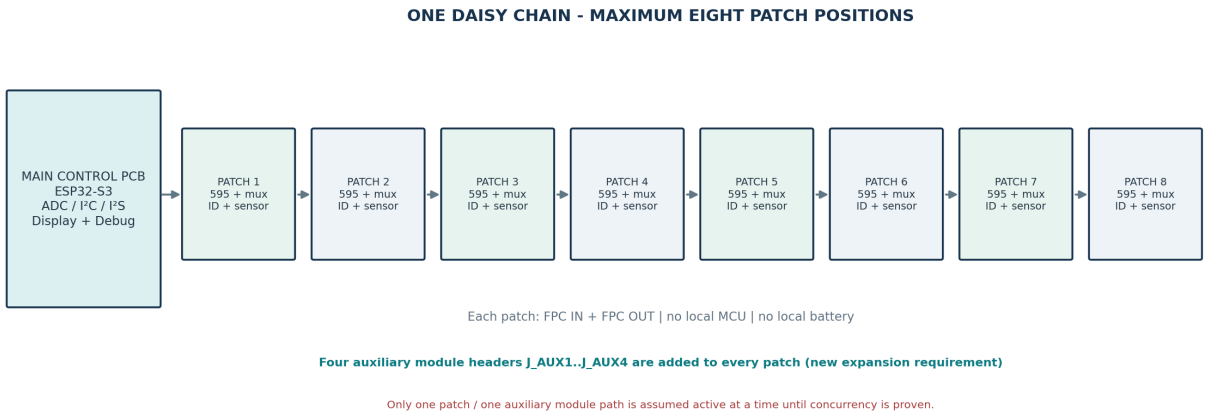
BEEBOX

V2 End-to-End Project Report

Modular honeybee-colony sensor platform | Electrical architecture | Expansion headers | Engineering workflow

PRODUCT INTENT

A modular sensor system for collecting hive-related environmental and acoustic measurements through one main controller and a daisy-chained set of batteryless sensor patches. The updated design adds four auxiliary module headers to every patch so users can attach selected external sensors, including methane, CO_2 and other compatible modules.



Document field	Value
Document ID	BBX-ENG-PR-003
Revision	0.2 - Expanded engineering baseline
Date	01 November 2026
Architecture	One ESP32-S3 main controller + up to 8 sensor patches
New change	4 auxiliary sensor-module headers per patch: J_AUX1 to J_AUX4
Status	Engineering proposal - conditional GO for prototype; not released for manufacture

IMPORTANT ENGINEERING STATUS

This report is an engineering baseline and implementation plan. It does not claim a built prototype, a connected EasyEDA API bridge, measured electrical performance, or manufacturing approval. Auxiliary-module compatibility depends on final sensor module part numbers and measured power/interface limits.

Document control and reading guide

This report extends the earlier Sensor Chain V2 feasibility proposal. It preserves the verified-in-document design intent and records the newly requested auxiliary sensor expansion as an engineering change.

Revision	Change summary	Status
0.1	Sensor Chain V2 feasibility baseline: one controller, up to eight MCU-free patches, 12-contact chain, T/RH or I²S microphone.	Reference baseline
0.2	Adds four auxiliary external-module headers to every patch and expands the control, switching, power, firmware and validation plan.	Current working baseline

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HOW TO USE THIS DOCUMENT

Use this as the project-level source of truth for the next schematic design cycle. Do not treat candidate component names, provisional auxiliary-header concepts or untested calculations as manufacturing-ready decisions. Freeze those items only after module datasheets, physical drawings and bench evidence are available.

01. Product purpose and operating concept

BeeBox V2 is a modular sensing platform intended to place different measurement functions at selected points around a bee colony / hive environment.

Primary purpose

Provide a main controller that can discover, select and read sensor functions distributed across up to eight daisy-chained sensor patches. The initial patch population supports temperature/humidity sensing or digital I²S audio sensing. The new expansion requirement adds four user-accessible headers per patch for optional external sensor modules, including gas and environmental modules such as methane and CO_2 sensors.

What the system is designed to do

- Collect environmental measurements from a selected temperature/humidity sensor patch.
- Capture audio from a selected digital I²S microphone patch.
- Allow optional external modules to be connected through four auxiliary headers on each patch, subject to an approved module compatibility list.
- Use one central controller to manage patch selection, module power, signal routing, identification and display/status functions.
- Scale physically to a maximum of eight patches on one chain, while retaining the same logical 12-contact connector interface for FPC and pogo implementations.

Design principles

- Centralized control: patches contain no local MCU and no battery; small logic, switching and power devices are allowed.
- One active path: activate and read one patch / one selected external module path at a time unless concurrent operation is separately proven.
- Modularity: use a repeatable patch PCB with clearly labelled ports and replaceable sensor modules.
- Fail-safe: all sensor power and signal paths default OFF during reset, shift-register updates and fault conditions.
- Evidence-led release: design calculations, ERC/DRC, netlist checks and physical measurements are required before fabrication release.

PRODUCT BOUNDARY

This document defines a sensing platform architecture. It does not claim validated bee-health diagnosis, colony-health scoring, methane interpretation, CO_2 interpretation, cloud analytics or field performance. Those are separate product and algorithm requirements.

02. Requirements and scope baseline

Requirements below combine the established Sensor Chain V2 baseline with the new auxiliary-header change.

Area	Requirement for current design iteration
Main board	One ESP32-S3 main controller PCB, one 12-contact sensor-chain connector, separate display interface and independent debug/programming access.
Patch count and topology	Maximum eight patches in one physical daisy chain: Main → Patch 1 → ... → Patch 8.
Patch electronics	No local MCU and no patch battery. Shift-register, analog switch/mux, ID and load-switch circuitry is permitted.
Primary sensor population	One primary type per patch: temperature/humidity OR digital I²S microphone.
Auxiliary module expansion - NEW	Four additional user-accessible headers on every sensor patch, labelled J_AUX1, J_AUX2, J_AUX3 and J_AUX4. Intended for selected external modules such as methane, CO_2 and other compatible sensors.
Auxiliary concurrency	One selected auxiliary port/module path at a time is the safe starting assumption. Four connected modules do not imply four simultaneous active channels.
Patch connections	Every patch has 12-way FPC IN and 12-way FPC OUT. Pin 8 is directional through the shift-register cascade, not a copper pass-through.
Power	The +3V3_CHAIN rail powers always-on patch logic. Primary sensor power is locally switched. Auxiliary module power requires a separate budget and may need 5 V generation or another distribution decision.
Display	Separate 4-wire I²C display connector: GND, +3V3_DISPLAY, DISPLAY_SCL and DISPLAY_SDA.
Alternate connection	Pogo-pin build keeps the same logical 12-signal interface as the FPC chain.

SCOPE CHANGE CONTROL

The four auxiliary headers are now part of the project requirement. The earlier “no third sensor class” limitation is superseded for expansion planning. However, no specific gas sensor module is approved yet; electrical compatibility is not universal and must be proven per module MPN.

03. System architecture

The backbone remains a centralized controller and a serial patch chain. Auxiliary module ports are added locally at each patch; they do not automatically increase the number of FPC chain contacts.

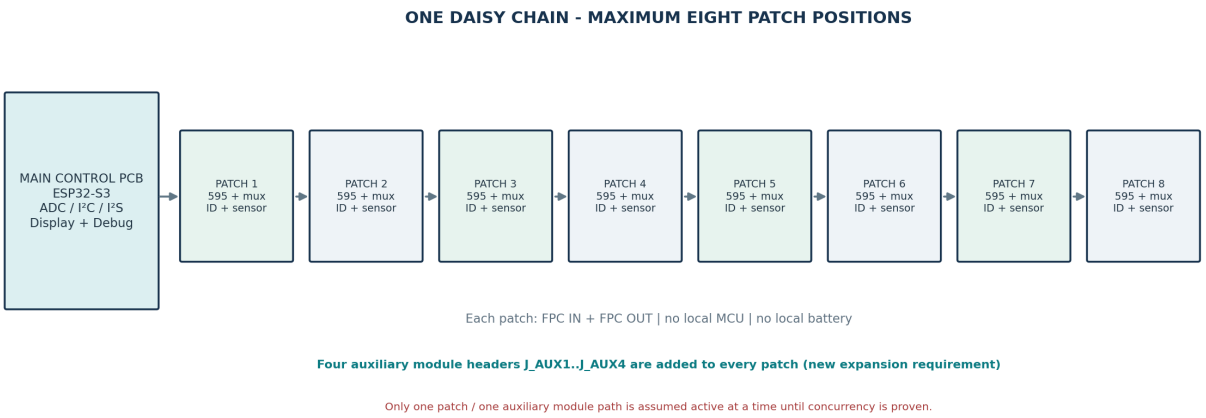


Figure 1. Updated logical architecture. Each patch has four auxiliary ports in addition to its primary sensor population.

Functional partition

Block	Responsibilities
Main controller PCB	ESP32-S3 runtime control; 64-bit shift-register control image; patch scanning; ADC identification; sensor bus mode selection; display/status; power supervision and debug.
12-contact chain	Always-on patch logic supply, common grounds, three time-multiplexed sensor/data lines and four shift-register control/cascade signals.
Sensor patch base	FPC IN/OUT; one SN74HC595-class control register; local sensor power switch; signal isolation; patch type ID; primary T/RH or I²S sensor footprints.
Auxiliary expansion	Four module headers, module-power control, input protection/conditioning, port selection/multiplexing and a future module identification strategy.
Firmware and tools	Safe state machine; sensor drivers; module profiles; project-controlled EasyEDA scripts; independent validation records.

IMPORTANT ARCHITECTURAL RULE

Do not wire all four auxiliary headers directly in parallel to the same signal bus. A selected-port switch/multiplexer is required to prevent bus contention, output-driver collisions and accidental powering through signal pins.

04. Main controller design

The main board coordinates the complete chain and keeps the display bus separate from the sensor-chain bus.

Core controller

The baseline calls for an ESP32-S3 main controller. The exact module MPN, flash/PSRAM option and GPIO map must be frozen from the selected module datasheet and the final PCB pin allocation. The shared sensor-data GPIO is intended to serve as the ADC input during identification and I²S DIN during microphone capture; that alternate peripheral use is a critical bench-validation item.

Interface	Baseline function / design note
Patch control	Four GPIOs: PATCH_SHIFT_DATA, PATCH_SHIFT_CLK, PATCH_LATCH and PATCH_OE_N.
Sensor clock/sync	Two shared GPIOs: I²C SCL/SDA in T/RH mode, or I²S BCLK/WS in microphone mode.
Sensor data / ID	One shared data path: ADC ID measurement during discovery; I²S SD/DIN during microphone capture. ADC and I²S modes are mutually exclusive.
Display	Separate 4-wire I²C connector; do not combine display pull-ups with the shared sensor bus.
Debug	Keep USB or UART/JTAG and test pads independent of the sensor chain.
Power supervision	Measure or monitor main regulated rail and battery state as selected in the power design. Exact telemetry and battery gauge are TBD.

Main-board design gates

- Reserve an ESP32-S3 ADC1-capable GPIO for the ID/data function; the previous repository draft GPIO38 assignment is not acceptable as the ADC pin in the feasibility baseline.
- Confirm that the same selected pad supports I²S DIN under the chosen module and firmware configuration.
- Confirm the GPIO output levels, bootstrapping constraints, USB/debug needs and antenna keep-out for the exact module.
- Keep display pull-ups and display power independent from the sensor-chain bus.
- Do not freeze the main PCB power stage until the battery range, maximum patch load and auxiliary-module power plan are calculated.

05. Twelve-contact sensor-chain interface

This is the current logical interface baseline. The previous repository pin assignment must be replaced consistently in schematics, netlists, PCB, firmware and documentation.

Pin	Net name	Function	Routing rule
1	GND	Ground / return	IN-to-OUT pass-through
2	+3V3_CHAIN	Always-on 3.3 V for patch logic and switched sensor input	IN-to-OUT pass-through
3	GND	Ground / return	IN-to-OUT pass-through
4	SENSOR_CLK	I²C SCL in T/RH mode; I²S BCLK in mic mode	IN-to-OUT pass-through
5	SENSOR_SYNC	I²C SDA in T/RH mode; I²S WS in mic mode	IN-to-OUT pass-through
6	GND	Ground / signal return	IN-to-OUT pass-through
7	SENSOR_DATA_ID	I²S SD during mic capture; ADC ID voltage during discovery	IN-to-OUT pass-through
8	PATCH_SHIFT_DATA	Shift data enters local DS; local QH′/Q7S exits to next patch	Directional through 74HC595
9	PATCH_SHIFT_CLK	Shared shift-register clock	IN-to-OUT pass-through
10	PATCH_LATCH	Shared storage latch	IN-to-OUT pass-through
11	PATCH_OE_N	Global active-low output enable	IN-to-OUT pass-through
12	GND	Ground / return	IN-to-OUT pass-through

PIN 8 IS NOT A NORMAL PASS-THROUGH

The upstream pin 8 signal connects to the local shift-register serial data input. The local serial output QH′/Q7S drives pin 8 of the downstream OUT connector. The other listed nets pass through. Verify connector pin numbering and mirror orientation using the exact manufacturer drawings before PCB layout.

Physical connector candidate

The earlier feasibility baseline lists Hirose FH34SRJ-12S-0.5SH(50) as a candidate 12-position, 0.5 mm-pitch ZIF receptacle for a 0.3 mm FPC/FFC. It is not yet a released footprint: verify the exact product drawing, land pattern, contact side, insertion direction, stiffener, flex bend and current margin. Pogo contacts are a separate mechanical implementation with the same logical pinout. [R2]

06. Universal sensor-patch base design

The patch is a passive controlled node: the main board addresses it through the cascade and selects the local sensor path.

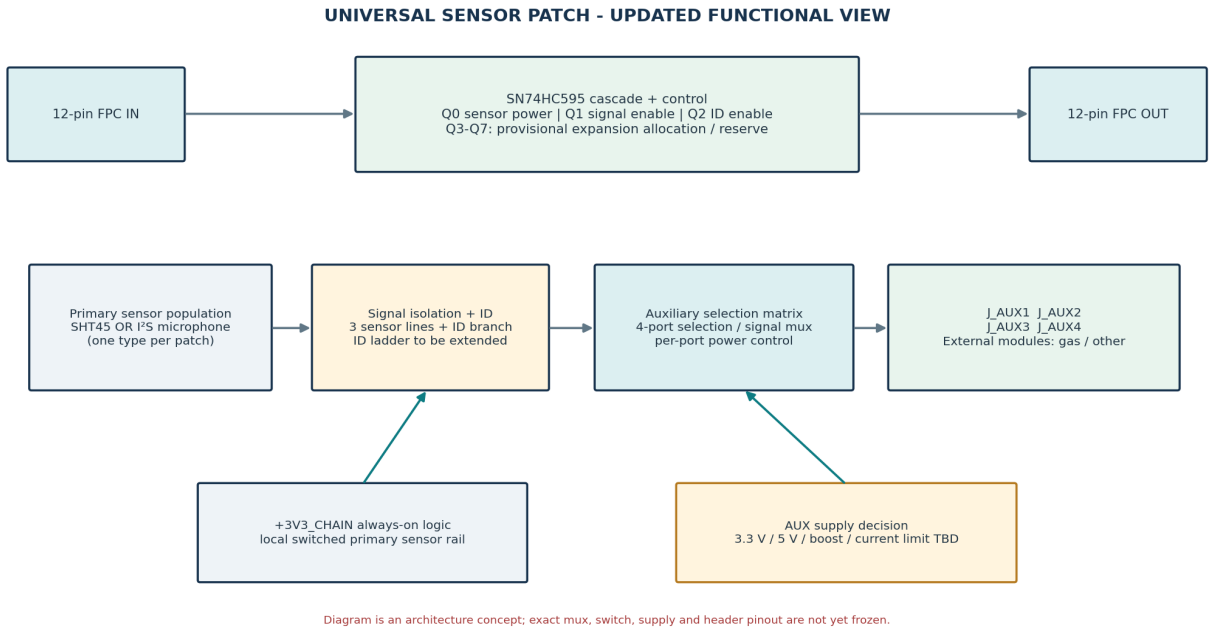


Figure 2. Patch functional view including the new auxiliary expansion subsystem.

Circuit block	Baseline / required function
FPC IN / OUT	Two 12-position connectors; pass through all shared rails and bus nets except directional shift data on pin 8.
Shift register	One SN74HC595-class 8-bit register per patch. Q0 sensor power, Q1 primary sensor signal enable, Q2 patch ID enable. Q3-Q7 are now under review for auxiliary selection/control.
Primary sensor isolation	Three signal paths to the selected sensor plus one switched ID-resistor branch. A CD74HC4066-class device is the earlier concept candidate; verify on-resistance, bandwidth, logic thresholds and signal integrity.
Primary sensor power	Local load switch creates the sensor VDD from +3V3_CHAIN. Defined default-off state and discharge behavior required.
Primary sensor population	One type per patch: SHT45-class T/RH OR a qualified I²S microphone. Do not populate both as live drivers unless the switching design proves safe isolation.
ID resistor	Earlier proposal: 22 kΩ for T/RH and 68 kΩ for mic, selected only when that patch is being identified. Recalculate / extend for aux modules.
Aux headers	Four external module ports J_AUX1..J_AUX4 on every patch; each is separately identified in the schematic and silkscreen.

PATCH POPULATION / ASSEMBLY RULE

For the primary sensor footprints, exactly one of T/RH or I²S mic is populated per patch in the initial architecture. Any alternate unpopulated branch must be DNI or isolated so that it cannot load or drive the shared sensor bus.

07. New requirement: four auxiliary headers per patch

This is the major revision in this report. It allows users to attach selected external modules without changing the basic 12-contact chain connector at this stage.

REQUIREMENT CAPTURED

Every sensor patch shall include four additional auxiliary module headers, with reference designators J_AUX1, J_AUX2, J_AUX3 and J_AUX4. The intent is to connect user-selected modules such as methane, CO_2 and other compatible sensors.

Intended capability

The four headers are expansion ports, not four new chain contacts. They are physically located on each sensor patch and connect through a patch-local power and signal selection subsystem. The same patch still connects to the main board using the 12-contact FPC chain. The first implementation should select one auxiliary port at a time, power it only when needed, route its signal(s) to the main controller, collect a reading and then return the port to a safe inactive state.

Proposed logical organization

Port	Purpose	Electrical status
J_AUX1	User plug-in module port 1	Header required; exact connector/pinout TBD
J_AUX2	User plug-in module port 2	Header required; exact connector/pinout TBD
J_AUX3	User plug-in module port 3	Header required; exact connector/pinout TBD
J_AUX4	User plug-in module port 4	Header required; exact connector/pinout TBD

Examples of intended sensor module classes

- Gas modules: methane / combustible-gas sensing modules, depending on the chosen module technology and calibration method.
- CO_2 modules: NDIR or other digital/analog CO_2 modules, depending on power and interface needs.
- Other environmental modules: only after their power, signal protocol, voltage range, warm-up time and environmental suitability are assessed.

What this does not yet promise

The addition of four headers does not mean every module from every vendor is plug-and-play. A methane or CO_2 module may use an analog output, I^2C, UART, PWM, SPI or another interface; it may require 3.3 V, 5 V or a higher supply; and its heater / sensing element may draw far more current than the original T/RH or microphone sensor. The project must define an approved module compatibility list.

SAFE INITIAL OPERATING ASSUMPTION

One auxiliary module port is selected and measured at a time. Concurrent reading of multiple auxiliary modules is a future capability until the mux, power budget and bus protocol are validated.

08. Auxiliary header electrical design and open gates

The headers must be designed around real sensor modules. The following is the engineering plan, not a frozen electrical pinout.

Recommended connector concept

A compact 4-position header is a reasonable starting concept for many modules that use supply, ground and two signal contacts: V_AUX, GND, AUX_IO_A and AUX_IO_B. This is a proposal only. Select the actual pin count and keying after reviewing the required methane/CO_2 module datasheets. Some modules need more signals, a separate analog output, enable, calibration or heater connections and may require a larger connector or a small module adapter.

Interface type	Possible use of two signal pins	Required design provisions
Analog output	One signal pin carries analog output; second may be unused / mode pin.	Input divider / buffer / filter as required; clamp and ADC range protection. Never connect a 5 V analog output directly to an ESP32-S3 GPIO.
I^2C	AUX_IO_A = SCL; AUX_IO_B = SDA.	Level compatibility, pull-up placement, address conflicts, bus capacitance and selected-port isolation.
UART	AUX_IO_A / AUX_IO_B as RX/TX by defined direction.	Crossed TX/RX convention, logic voltage matching, baud rate and optional level translation.
PWM / digital	One signal can carry PWM or logic state.	Input conditioning, level compatibility and firmware capture method.
SPI / multi-wire	Two signal pins are generally insufficient for a full SPI bus.	Use a larger connector or dedicated adapter; do not claim SPI support on the 4-contact concept.

Selection, isolation and identification

- Provide a port-selection network so inactive headers do not load the shared bus and two outputs cannot drive one line simultaneously.
- Provide controlled module power. Prefer a selected-port power path with a load switch and a safe default-off state.
- Determine whether the existing SN74HC595 outputs Q3-Q7 can implement the required selection or whether a second register, decoder or dedicated mux controller is needed.
- Extend the patch identification strategy. Existing 22 kΩ / 68 kΩ codes only distinguish the primary T/RH and microphone classes; they do not identify arbitrary aux module types or four separate ports.
- Add ESD / transient protection and overvoltage protection based on the external connector exposure and module output range.

POWER WARNING FOR GAS MODULES

Do not assume that +3V3_CHAIN can power any methane or CO_2 module. Some modules require 5 V or have heater / optical-source current demands. A 5 V module may need a dedicated boost rail, extra regulation, current limiting or an external supply strategy. Calculate worst-case current through the chain before approving a module.

09. Identification model and firmware operating sequence

The controller identifies position through the serial shift-register order and identifies sensor type through a switched ID path or a future module profile mechanism.

Existing primary sensor ID proposal

Patch population	R_ID to +3V3_CHAIN	Main pull-down	Nominal ADC voltage
No sensor / open	Open	100 kΩ to GND	~0 V
T/RH	22 kΩ, 1%	100 kΩ, 1%	~2.705 V
I²S microphone	68 kΩ, 1%	100 kΩ, 1%	~1.964 V

The prior feasibility calculation uses V_ID = V_CHAIN × R_PD / (R_ID + R_PD). Initial windows were: below 0.25 V = empty; 1.70-2.20 V = microphone; 2.50-2.95 V = T/RH; all other values = invalid. These windows are screening values and require calibration, tolerance analysis and ADC tests. They are not suitable as a finished codebook for four new auxiliary ports.

Updated firmware state machine

- SAFE_BOOT: hold PATCH_OE_N high; all outputs disabled; verify chain-control GPIO levels.
- CHAIN_INIT: shift a known zero image through all eight registers; latch; verify bit order using walking-one / walking-zero tests.
- PATCH_DISCOVERY: identify one patch position at a time; select its primary ID path and classify the known primary type.
- AUX_DISCOVERY: select J_AUX1, J_AUX2, J_AUX3 or J_AUX4 individually; apply a future approved ID/configuration method and log unknown / absent modules safely.
- CONFIGURE: select protocol profile (I²C, I²S, UART, analog or supported digital mode), establish pull-ups/levels and sensor power.
- ACQUIRE: wait for module-specific warm-up / startup; collect data and timestamps; perform plausibility and fault checks.
- ISOLATE: disconnect signal path first, disable module power, discharge if required, then advance to next port or patch.
- FAULT: leave the failing path off, record patch position / port / reason, continue only if safe.

CRITICAL PIN REUSE CHECK

The shared SENSOR_DATA_ID GPIO must work as an ADC input during identification and I²S DIN during audio capture on the exact ESP32-S3 module. Prove this on the bench. If it fails, revise the main-board GPIO or add an analog mux / dedicated input.

10. Power, mechanical and PCB design

The existing chain rail supports the always-on control logic. The auxiliary modules introduce a new load class that must be budgeted explicitly.

Power architecture

Rail / block	Purpose	Design decision
Battery input	Baseline 1-cell LiPo supply.	Cell, protection, charger, connector and safe cutoff to be confirmed.
Main 3.3 V	ESP32-S3 and main logic.	Regulator must cover controller peaks and other board loads.
+3V3_CHAIN	Always-on patch logic and input to local sensor power switches.	Budget eight registers, switches, pull-ups, leakage and cable drop.
Primary sensor VDD	One selected T/RH or mic sensor at a time.	Local patch load switch; default OFF; confirm startup and discharge.
Auxiliary module V_AUX	User-selected modules on J_AUX1..4.	Voltage options, boost/regulation, current limit and individual port switching TBD from chosen module list.
Display rail	Separate display supply / I²C interface.	Verify module voltage, address and onboard pull-ups.

Important baseline regulator note

The earlier project baseline includes a 1-cell LiPo and a TPS62162-class buck concept. A buck-only regulator cannot guarantee a regulated 3.3 V output over the full cell discharge range when input headroom falls below the required level. Before freezing V2, choose a suitable buck-boost solution or define and validate a battery cutoff that maintains the rail.

PCB and mechanical rules

- Keep the FPC connector orientation, pin-1 marking and IN/OUT silkscreen unmistakable; validate with a 1:1 print and continuity fixture.
- Keep gas-sensor openings, acoustic ports and temperature/humidity sensing area clear of enclosure blockage and heat sources.
- Review condensation, dust, propolis, vibration, flex bend radius, connector strain relief and pogo compression / cycle life.
- Place local decoupling close to IC supply pins; tune sensor bypass capacitance from the selected datasheets and inrush analysis.
- Route high-impedance analog measurements away from switching power and fast digital lines; provide ground return continuity.
- Use small footprints where electrically and mechanically appropriate; do not force every passive into a tiny size. Power capacitance, voltage rating, tolerance, derating and assembly yield take priority.

FOOTPRINT NOTE

When specifying small passives, distinguish metric and imperial package codes. A 0404 metric component is not the same physical size as a 0402 imperial component. Freeze every footprint against the actual manufacturer package drawing.

11. Software tools and API integration workflow

The development workflow uses EasyEDA Pro as the schematic authority, with terminal/agent tools for controlled automation and review.

Tool / environment	Current project role / observed status
EasyEDA Pro	Schematic and PCB design environment. Screenshot shows EasyEDA Pro V3.2.149 with a new BEE BOX V2 project and an empty schematic sheet ready for structured capture.
EasyEDA API Gateway	Gateway v1.0.6 is visible in the application, but the screenshot showed “Bridge not found, retrying”. Connection is NOT yet verified. First task is a bridge handshake / smoke test, without modifying the schematic.
Herdr	Windows terminal workspace manager. BeeBox V2 workspace created successfully with workspace ID WA; one tab and one root pane were reported.
Firstmate / Pi agent	Agent runtime is visible in the BeeBox V2 Herdr workspace. Use it for task orchestration, controlled file/process interaction and engineering review.
PyAgent / local validation	Use where configured for script checks, pin-map comparisons, report generation and repeatable validation; exact integration and commands must be confirmed in the installed environment.
JavaScript scripts	Use small, subsystem-specific EasyEDA scripts rather than one large generator.

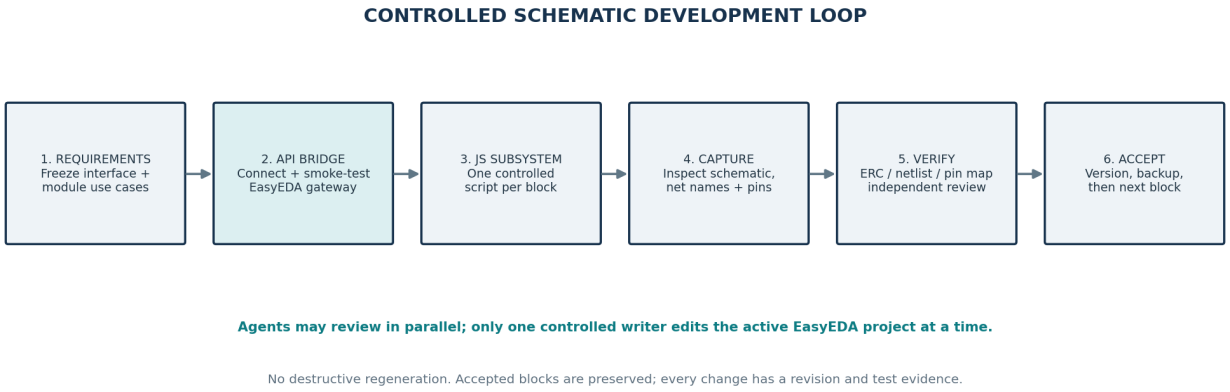


Figure 3. Recommended controlled design loop.

API BRIDGE FIRST TASK

Before asking an agent to create schematic objects, confirm that the API bridge is discovered, the gateway handshake completes, a harmless read-only query succeeds and the project/sheet context is correct. Do not start schematic generation while the bridge is retrying.

12. Schematic capture and JavaScript methodology

Build the design in reviewable increments. Each script must target a named subsystem and preserve all previously accepted work.

Recommended schematic sheet plan

Board	Sheet / block	Scope
Main	M01 Power input	Battery connector, protection, charge interface if selected, regulator(s), power test points.
Main	M02 ESP32-S3 core	Module, boot/reset, programming/debug, decoupling and GPIO map.
Main	M03 Chain control + ADC	Shift data/clock/latch/OE, ID ADC, safe-state pull resistors and test points.
Main	M04 12-pin FPC / pogo	Exact 12-pin map, connector orientation, labels and continuity points.
Main	M05 Display + status	Independent 4-wire I²C display and status LED/test points.
Patch	P01 FPC IN / OUT	Two connectors, pin-8 directional cascade, pass-through nets.
Patch	P02 Register + power	74HC595 control, defaults, local load switch and decoupling.
Patch	P03 Primary sensor / ID	4066-class isolation, ID resistor branch, T/RH / microphone population variants.
Patch	P04 AUX1-AUX4	Four module headers, port mux/switch matrix, power selection, protection and pin labels.
Patch	P05 Test / assembly	Test points, DNI options, variant notes, configuration and manufacturing labels.

Rules for every JS script

- Unique script filename, subsystem ID, revision, target project/sheet and expected component references.
- Use exact library symbols, device mapping, package and manufacturer part number only after validation.
- Create explicit nets and actual electrical wires; labels alone must not hide missing connections.
- Include pin-to-net map, placement coordinates, orientation, values, footprints and test points.
- Avoid destructive clear/delete operations; scripts must be additive or narrowly scoped and safe to rerun where possible.
- After execution, inspect the sheet visually, verify symbol pins and nets, export / compare netlist, run ERC and record the result.
- Only after user acceptance and a saved project revision should the next subsystem script be prepared.

13. Component selection and BOM rules

Select components from electrical requirements first, then confirm exact availability, package, lifecycle and footprint mapping.

Subsystem	Candidate / baseline concept	Qualification before selection
Main MCU	ESP32-S3 module	Exact MPN, memory, antenna / keep-out, GPIO capability, ADC/I²S alternate use, supply peaks and lifecycle.
T/RH	Sensirion SHT45	Supply, accuracy, port placement, decoupling, humidity exposure, package / footprint. [R8]
I²S microphone	Current-production digital I²S mic; PUI DMM-4026-B-I²S family is one candidate from prior study.	Lifecycle, BCLK limits, output format, acoustic port, sample rate and full-chain timing. Earlier ICS-43434 baseline is marked EOL. [R9-R11]
Shift register	SN74HC595	Supply range, output current, clock, serial output pin, SRCLR/OE behavior, exact package. [R3]
Signal switching	CD74HC4066-class quad switch	On-resistance, bandwidth, signal range, logic threshold, off leakage, I²S waveform and power-off isolation. [R4]
Load switch	TPS22918-class or qualified alternate	Input voltage, current, turn-on behavior, reverse blocking / backfeed, discharge and default-off control. [R5]
AUX signal mux	To be selected after module shortlist	Number of channels, analog range, R_ON, bandwidth, leakage, signal direction, voltage domain and control logic.
AUX power	Possible dedicated boost / regulated rail	Per-module voltage/current, heater/inrush, thermal dissipation, chain current, protection and run-time impact.
Connectors	12-way FPC candidate + four auxiliary headers per patch	Exact manufacturer drawing, mating parts, pitch, retention, current, orientation and mechanical access.

BOM acceptance rules

- Every line must have a unique reference designator, manufacturer, exact MPN, quantity, value, tolerance, voltage/current rating, package and footprint.
- Verify symbol-to-footprint pin mapping and pad numbering against the manufacturer drawing. Check lifecycle and at least one alternate for critical parts.
- Use 0404 metric passives as a compact target only where suitable; choose larger packages when needed for capacitance, voltage, current, power, derating or assembly reliability.
- For external modules, record module MPN, supply voltage, maximum/average current, protocol, logic level, warm-up time, calibration needs and output limits.

14. Verification and acceptance plan

A clean-looking schematic or successful script execution is not evidence of electrical correctness. Each gate needs a reproducible artifact.

ID	Test	Acceptance evidence
V-01	Safe power-up	OE_N disables outputs; sensor and auxiliary supplies remain OFF until selected. Scope + current log.
V-02	64-bit chain order	Walking-one / walking-zero test maps all eight patches and verifies pin-8 direction. Logic-analyser log.
V-03	Primary patch ID	T/RH / mic / empty values classify correctly over tolerance and rail variation. ADC CSV / calibration table.
V-04	Auxiliary port select	J_AUX1..4 are individually selected; inactive ports cannot drive or load shared data lines. Test report.
V-05	Auxiliary module power	Correct selected rail and current limit at each header; inactive port off; no back-power. DMM / current trace.
V-06	Auxiliary input range	Analog and digital signal levels stay within GPIO / ADC limits; protection response recorded. Scope + calculations.
V-07	I²C T/RH	Only selected SHT45 responds; bus rise time and sink current pass. I²C capture.
V-08	I²S microphone	Only selected microphone drives SD; capture works at patch 1 and patch 8. Audio file + scope.
V-09	Aux protocol profiles	Each approved module MPN communicates with its defined protocol and expected response. Raw logs.
V-10	Power-off backfeed	Inactive module rail and I/O protection paths do not unintentionally energize the sensor. Current / voltage log.
V-11	Full-chain power / SI	Patch-8 rail, I²C timing, I²S edge quality and module startup pass at maximum intended chain length. Scope.
V-12	FPC / pogo map	All 12 logical contacts, orientation and pin-8 cascade pass automated continuity fixture.
V-13	Fault recovery	Missing module, invalid ID, timeout, reset and open-chain conditions leave all outputs safe. Firmware log.
V-14	Mechanical / thermal	No connector lift, flex damage, blocked sensor port, unacceptable heating or enclosure interference. Inspection record.

FIRST PROTOTYPE STOP-GATE

Build one main controller and two patches first: one T/RH patch and one I²S mic patch. Include an AUX proof on at least one patch using representative external module(s). Do not scale to eight patches until chain order, ID, primary isolation, aux port switching, module power and back-power tests pass.

15. Risk register and engineering stop-gates

The auxiliary ports introduce new high-priority risks and do not remove the original V2 electrical risks.

Risk	Impact	Mitigation / closure evidence	Priority
Aux module voltage/current unknown	Header may damage module or brown out chain.	Select representative module MPNs; compute peak/inrush; define V_AUX and current limit; test patch 8.	Critical
Four ports tied together	Bus contention, false readings or back-power.	One-of-four mux / switch matrix; verify off isolation and port-by-port operation.	Critical
Analog output exceeds ADC range	ESP32-S3 damage or incorrect readings.	Input divider / buffer / clamp; tolerance analysis; validate maximum fault voltage.	Critical
Aux protocol diversity	Some modules cannot use two shared signal lines.	Approved compatibility matrix; larger header / adapter for unsupported protocols.	High
Shift-register outputs insufficient	Control architecture cannot select/power all ports safely.	Reallocate Q3-Q7 only after control budget; add register/decoder if needed.	High
Aux module ID not defined	Firmware cannot identify which module/port is present.	Define resistor ID / EEPROM / configuration profile per port; unknown modules fail safe.	High
5 V or heater modules on 3.3 V chain	Rail collapse, heat or excess battery drain.	Separate boost / power strategy; one-at-a-time power; calculate thermal and chain current.	Critical
Shared ADC / I²S GPIO fails	Patch ID or mic capture cannot work as proposed.	Bench test exact ESP32-S3 module; add mux or revise GPIO if necessary.	Critical
Mic lifecycle gap	BOM obsolescence.	Qualify current-production I²S mic and alternate.	High
Long-chain signal degradation	Mic or I²C communication unreliable.	Test full eight-position physical chain; scope at patch 1 and patch 8.	Critical
1S LiPo buck headroom	3.3 V chain rail drops at low battery.	Buck-boost design or validated early cutoff; measure under load.	High
Connector orientation mirrored	Power or data pins reversed.	Manufacturer drawing review, pin-1 marks and 1:1 continuity fixture.	Critical
Mid-chain patch removal	Downstream patches become unreachable.	Document service limitation; bypass only if separately required.	Medium

16. Execution phases and deliverables

Recommended work order for efficient use of EasyEDA scripts and agent/model time.

Phase	Work package	Exit deliverable / gate
P0 - Baseline update	Approve four headers per patch; select target module families; define patch outline, FPC lengths, battery and power envelope.	Signed requirements + interface control document; open questions tracked.
P1 - Module survey	Collect 2-4 representative methane / CO_2 / environmental module datasheets and exact MPNs. Build protocol, supply and current matrix.	Approved module shortlist and port compatibility table.
P2 - Architecture closure	Allocate auxiliary connector pins, supply strategy, mux/switch topology, ID model and shift-register controls.	Updated architecture, GPIO / control map, power budget and net dictionary.
P3 - API / tooling setup	Connect EasyEDA API Gateway; run read-only bridge smoke tests; confirm library/API schema and project target.	Verified bridge handshake and safe API test report.
P4 - Schematic capture	Create main-board sheets and patch sheets as separate JS subsystems. Add auxiliary headers and selection/power circuitry.	Complete schematic blocks, component list, netlist and pin map.
P5 - Independent audit	Check symbol/device/footprint mapping, wire connectivity, ERC, netlist parity and visual organization.	Audit report; no unresolved critical pin / net mismatch.
P6 - Two-patch proof	Fabricate / assemble controller + two patches; test one T/RH, one mic and at least one AUX module path.	Measured prototype report; Gate 6 pass.
P7 - Eight-position validation	Assemble eight patches; test mixed population, missing tail, all port selection, maximum cable and worst-case power.	Eight-patch validation matrix with raw evidence.
P8 - Controlled release	Close critical/high risks; finalize BOM, Gerbers, drill, PnP, assembly drawings, firmware and test fixture.	Manufacturing release package with revision/change control.

Indicative schedule from the prior feasibility baseline

The earlier baseline estimated short design phases for requirements, feasibility, schematic/proof-board, and a two-patch prototype, followed by a further layout, firmware and eight-patch validation cycle. Those estimates do not include fabrication, supplier lead times, customs, assembly queues or redesign. The new auxiliary sensor requirement adds module-selection, power and mux design work; schedule must be re-estimated after the module shortlist is known.

17. Project files, working method and release package

Keep all engineering artifacts in the BeeBox V2 project folder and align every update to the same approved revision.

Recommended deliverable structure

Folder / artifact	Contents
docs/	Project report, requirements, architecture, component research, BOM, power design, interface guide, schematic plan, risk register, validation plan and revision history.
schematic/	Main-controller and sensor-patch EasyEDA schematics; auxiliary module interface variants.
easyeda_scripts/	One JavaScript file per subsystem, with metadata, revision and safe update behavior.
netlist/	Canonical exported netlists, pin-to-net CSVs, connector maps and comparison output.
bom/	Controlled BOM, alternates, manufacturer drawings, module compatibility matrix and sourcing notes.
pcb/	PCB source, Gerbers, drill, stack-up, board outlines, fabrication notes, pick-and-place and assembly drawings.
firmware/	Patch scan state machine, drivers, module profiles, ADC calibration, logs and test builds.
verification/	ERC/DRC results, scope captures, logic analyser logs, power/current records, calibration and acceptance tests.
release/	Only approved manufacturing package, firmware version, test fixture spec and change log.

Release package must include

- Editable EasyEDA project and exact library component/device mappings.
- PDF schematic exports, board layout, Gerbers, drill files, pick-and-place, fabrication / assembly drawings and stack-up.
- BOM with exact manufacturer part numbers, packages, alternates and lifecycle status.
- Canonical netlist plus connector pin map and pin-to-pad verification report.
- ERC / DRC reports, design review sign-off, firmware build and version, test procedure and measured validation evidence.
- Four auxiliary header pinout / voltage labels, supported module list, max current per port and safety warnings.

WORKSPACE NOTE

Herdr reports a BeeBox V2 workspace (ID WA) in the project folder. EasyEDA Pro is open on the new project, but the API Gateway bridge was still retrying in the provided screenshot. The bridge is a setup gate, not a completed integration.

18. Open decisions, assumptions and references

Close these inputs before the schematic and PCB are treated as a frozen design.

Open engineering decisions

Decision	What is needed to close it
Four-header interpretation	This report assumes four AUX connectors on every patch. Confirm the physical count is per patch and keep J_AUX1..J_AUX4 on each patch.
Aux module set	Exact methane, CO_2 and other sensor module MPNs, datasheets and preferred vendors.
Aux connector geometry	Pin count, pitch, locking/keying, through-hole vs SMT, cable/board mating and user insertion cycles.
Aux interface pins	Voltage, GND, signal count, analog input, I^2C/UART mapping, module detect / ID method and whether adapters are needed.
Aux power	3.3 V / 5 V / other supply, current per port, max simultaneous current, boost/regulator and thermal limit.
Aux concurrency	One port at a time for first build; decide whether multi-port concurrency is a future requirement.
Aux identification	Resistor code, EEPROM, manual firmware profile or another discovery mechanism.
Control expansion	Whether Q3-Q7 suffice or a second shift register / mux decoder is required.
Primary mic	Current-production I^2S MPN, sample rate, bit depth, BCLK and acoustic port.
Battery / regulator	Cell and protection MPN, charge path, cutoff, 3.3 V regulator and optional 5 V aux rail.
Mechanical details	Patch dimensions, FPC segment length, overall chain length, enclosure, gas aperture, acoustic aperture and environmental protection.
Pogo implementation	Exact contact MPN, geometry, compression, retention and service-cycle target.
Display	Exact module MPN, address, voltage and pull-up presence.

Engineering references carried forward from the feasibility baseline

[R1] BeeBox repository and V1/V2 source documents: https://github.com/NijoP/BeeBox
[R2] Hirose FH34 series product information: https://www.hirose.com/en/product/p/CL0580-1253-0-50
[R3] Texas Instruments SN74HC595: https://www.ti.com/product/SN74HC595
[R4] Texas Instruments CD74HC4066: https://www.ti.com/product/CD74HC4066
[R5] Texas Instruments TPS22918: https://www.ti.com/product/TPS22918
[R6] Espressif ESP32-S3 ADC programming guide:
https://docs.espressif.com/projects/esp-idf/en/latest/esp32s3/api-reference/peripherals/adc/index.html
[R7] Espressif ESP32-S3 I^2S programming guide:
https://docs.espressif.com/projects/esp-idf/en/latest/esp32s3/api-reference/peripherals/i2s.html
[R8] Sensirion SHT45: https://sensirion.com/products/catalog/SHT45
[R9] TDK InvenSense ICS-43434 lifecycle/product page: https://www.invensense.tdk.com/en-us/products/microphone/ics-43434
[R10] PUI Audio DMM-4026-B-I2S-R: https://puiaudio.com/product/microphones/dmm-4026-b-i2s-r
[R11] PUI Audio DMM-4026-B-I2S evaluation document: https://api.puiaudio.com/filename/DMM-4026-B-I2S-EB-R.pdf
[R12] Mill-Max 0906-series catalog: https://www.mill-max.com/sites/default/files/external/catalog/2021-02/023-028.pdf

FINAL ENGINEERING STATUS

Conditional GO for schematic and two-patch proof work after the auxiliary-module interface is closed. NOT released for manufacture. No hardware test, full-chain signal-integrity proof, aux power qualification, API bridge handshake or environmental field trial is claimed by this document.

END OF DOCUMENT